

| 자동 저장 (끔)    ai_trend_dataset_template 검색                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |         |                        |                              |                            |                       |                          |                                 |                                    |                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------|------------------------|------------------------------|----------------------------|-----------------------|--------------------------|---------------------------------|------------------------------------|---------------------------|
| 파일 홈 삽입 그리기 페이지 레이아웃 수식 데이터 검토 보기 자동화 도움말                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |         |                        |                              |                            |                       |                          |                                 |                                    |                           |
| <div> <div> <div>잘라내기</div> <div>붙여넣기</div> <div>서식 복사</div> </div> <div> <div>맑은 고딕</div> <div>11</div> <div>가</div> <div>가</div> </div> <div> <div>가</div> <div>가</div> <div>가</div> <div>가</div> <div>가</div> <div>가</div> <div>가</div> <div>가</div> </div> <div> <div>자동 줄 바꿈</div> <div>병합하고 가운데 맞춤</div> </div> <div> <div>일반</div> <div>%</div> <div>0.00</div> <div>00</div> </div> <div> <div>표준</div> <div>나뭇잎</div> <div>보통</div> <div>좋은</div> </div> <div> <div>조건부 서식</div> <div>표 서식</div> </div> <div> <div>삽입</div> </div> </div> |            |         |                        |                              |                            |                       |                          |                                 |                                    |                           |
| D16 :  100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |         |                        |                              |                            |                       |                          |                                 |                                    |                           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | A          | B       | C                      | D                            | E                          | F                     | G                        | H                               | I                                  | J                         |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | year_month | country | occupation             | source_indeed_total_it_index | source_indeed_ai_job_share | source_so_ai_use_rate | source_so_avg_salary_usd | source_linkedin_ai_hiring_index | source_stanford_genai_demand_score | key_skills                |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Jan-23     | 미국      | AI-ML Engineer         | 100                          | 9.22                       | 81.9                  | 118679                   | 110.4                           | 56.1                               | Python, PyTorch, LLM, RAG |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Jan-23     | 미국      | Data Scientist/Analyst | 100                          | 6.77                       | 74.3                  | 66467                    | 128.1                           | 72                                 | Python, SQL, AWS          |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Jan-23     | 미국      | Software Developer     | 100                          | 6.92                       | 64.3                  | 56791                    | 108.3                           | 89.2                               | Python, SQL, AWS          |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Jan-23     | 미국      | Cloud/DevOps           | 100                          | 7                          | 82.6                  | 73262                    | 123.1                           | 79.8                               | Python, SQL, AWS          |
| 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Jan-23     | 한국      | AI-ML Engineer         | 100                          | 7.42                       | 80.8                  | 102048                   | 101.3                           | 92.9                               | Python, PyTorch, LLM, RAG |
| 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Jan-23     | 한국      | Data Scientist/Analyst | 100                          | 6.35                       | 82.9                  | 87174                    | 106.4                           | 67.5                               | Python, SQL, AWS          |
| 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Jan-23     | 한국      | Software Developer     | 100                          | 3.72                       | 61.3                  | 73722                    | 113.7                           | 51.8                               | Python, SQL, AWS          |
| 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Jan-23     | 한국      | Cloud/DevOps           | 100                          | 10.22                      | 73.4                  | 107902                   | 103.6                           | 81                                 | Python, SQL, AWS          |
| 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Jan-23     | 일본      | AI-ML Engineer         | 100                          | 7.03                       | 80.7                  | 99481                    | 127.5                           | 80.3                               | Python, PyTorch, LLM, RAG |
| 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Jan-23     | 일본      | Data Scientist/Analyst | 100                          | 10.99                      | 71.2                  | 76114                    | 111.3                           | 91.7                               | Python, SQL, AWS          |
| 12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Jan-23     | 일본      | Software Developer     | 100                          | 6.28                       | 71.6                  | 79578                    | 115.7                           | 71.2                               | Python, SQL, AWS          |
| 13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Jan-23     | 일본      | Cloud/DevOps           | 100                          | 9.49                       | 66.6                  | 50420                    | 103.5                           | 56.6                               | Python, SQL, AWS          |
| 14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Jan-23     | 독일      | AI-ML Engineer         | 100                          | 6.53                       | 81.4                  | 114101                   | 131.6                           | 51.3                               | Python, PyTorch, LLM, RAG |
| 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Jan-23     | 독일      | Data Scientist/Analyst | 100                          | 7.12                       | 71.3                  | 81157                    | 138                             | 70.8                               | Python, SQL, AWS          |
| 16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Jan-23     | 독일      | Software Developer     | 100                          | 3.52                       | 83.3                  | 104739                   | 102.4                           | 80.2                               | Python, SQL, AWS          |
| 17                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Jan-23     | 독일      | Cloud/DevOps           | 100                          | 3.53                       | 82.6                  | 129605                   | 132.9                           | 60.8                               | Python, SQL, AWS          |
| 18                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Jan-23     | 인도네시아   | AI-ML Engineer         | 100                          | 11.3                       | 81.4                  | 42871                    | 108.7                           | 81.8                               | Python, PyTorch, LLM, RAG |
| 19                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Jan-23     | 인도네시아   | Data Scientist/Analyst | 100                          | 3.8                        | 73.2                  | 107680                   | 112.4                           | 85.3                               | Python, SQL, AWS          |
| 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Jan-23     | 인도네시아   | Software Developer     | 100                          | 6.92                       | 63.2                  | 42751                    | 131.3                           | 65.5                               | Python, SQL, AWS          |
| 21                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Jan-23     | 인도네시아   | Cloud/DevOps           | 100                          | 7.62                       | 80.7                  | 64248                    | 123.3                           | 90.7                               | Python, SQL, AWS          |
| 22                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Jan-23     | 베트남     | AI-ML Engineer         | 100                          | 6.51                       | 82.5                  | 98037                    | 107.8                           | 83.7                               | Python, PyTorch, LLM, RAG |
| 23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Jan-23     | 베트남     | Data Scientist/Analyst | 100                          | 7.05                       | 75                    | 122888                   | 127.6                           | 91                                 | Python, SQL, AWS          |
| 24                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Jan-23     | 베트남     | Software Developer     | 100                          | 11.05                      | 80.7                  | 57317                    | 101.4                           | 91.1                               | Python, SQL, AWS          |
| 25                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Jan-23     | 베트남     | Cloud/DevOps           | 100                          | 6.82                       | 82                    | 124934                   | 126.6                           | 70.8                               | Python, SQL, AWS          |
| 26                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Feb-23     | 미국      | AI-ML Engineer         | 100                          | 3.68                       | 74.9                  | 79033                    | 138.1                           | 88                                 | Python, PyTorch, LLM, RAG |
| 27                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Feb-23     | 미국      | Data Scientist/Analyst | 100                          | 10.07                      | 75                    | 68901                    | 107.9                           | 84.3                               | Python, SQL, AWS          |
| 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Feb-23     | 미국      | Software Developer     | 100                          | 11.88                      | 82.5                  | 122957                   | 136.7                           | 67.4                               | Python, SQL, AWS          |
| 29                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Feb-23     | 미국      | Cloud/DevOps           | 100                          | 6.74                       | 78.5                  | 61217                    | 112.6                           | 62.4                               | Python, SQL, AWS          |
| 30                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Feb-23     | 한국      | AI-ML Engineer         | 100                          | 11.69                      | 83.9                  | 59996                    | 135.8                           | 58.5                               | Python, PyTorch, LLM, RAG |
| 31                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Feb-23     | 한국      | Data Scientist/Analyst | 100                          | 5.11                       | 82.3                  | 58624                    | 107                             | 54.3                               | Python, SQL, AWS          |
| 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Feb-23     | 한국      | Software Developer     | 100                          | 7.84                       | 84.8                  | 70869                    | 127.5                           | 92.4                               | Python, SQL, AWS          |

File Edit Selection View Go Run Terminal Help

data\_trend

ai\_trend\_visual.py

ai\_trend\_visual.py > create\_four\_page\_report

```
1 import matplotlib
2 matplotlib.use('Agg') # Headless 환경 설정
3 import matplotlib.pyplot as plt
4 from matplotlib.backends.backend_pdf import PdfPages
5 import seaborn as sns
6 import pandas as pd
7 import numpy as np
8 import os
9
10
11 def create_ai_trend_dashboard(data_path, output_image_path):
12     # 1. 폰트 및 스타일 설정 (한글 폰트 적용)
13     plt.rcParams['font.family'] = 'Noto Sans CJK JP'
14     plt.rcParams['axes.unicode_minus'] = False
15     sns.set_theme(style='whitegrid', font='Noto Sans CJK JP', palette='colorblind')
16
17
18 # 2. 데이터 로드 및 전처리
19 def _prepare_data(data_path):
20     if not os.path.exists(data_path):
21         raise FileNotFoundError(f"데이터 파일을 찾을 수 없습니다: {data_path}")
22
23     df = pd.read_csv(data_path)
24     df['datetime'] = pd.to_datetime(df['year_month'])
25
26     # 월별 시계열 데이터 집계
27     df_monthly = df.groupby('datetime')[['source_indeed_ai_job_share', 'source_so_ai_use_rate', 'source_linkedin_ai_hiring_index']].mean().reset_index()
28     df_monthly = df_monthly[['source_indeed_ai_job_share',
29                             'source_so_ai_use_rate',
30                             'source_linkedin_ai_hiring_index']]
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```

Problems Output Debug Console Terminal Ports

origin https://github.com/bigpark61/data\_trend.git (fetch)  
origin https://github.com/bigpark61/data\_trend.git (push)  
PS C:\Temp\data\_trend> git remote -v  
Enumerating objects: 5106, done.  
Counting objects: 100% (5106/5106), done.  
Delta compression using up to 6 threads  
Compressing objects: 100% (4639/4639), done.  
Writing objects: 100% (5106/5106), 41.89 MiB | 16.80 MiB/s, done.  
Total 5106 (delta 348), reused 5106 (delta 348), pack-reused 0 (from 0)  
remote: Resolving deltas: 100% (348/348), done.  
To https://github.com/bigpark61/data\_trend.git  
\* [new branch] main -> main  
branch 'main' set up to track 'origin/main'.  
PS C:\Temp\data\_trend>

Explorer

Open Editors

ai\_trend\_visual.py

data\_trend

.venv

Lib

Scripts

share

.gitignore

.lock

CACHEDIR.TAG

pyvenv.cfg

output

ai\_trend\_report\_4pages.pdf

ai\_trend\_visualization.png

templates

ai\_trend\_dataset\_template.csv

ai\_trend\_visual.py

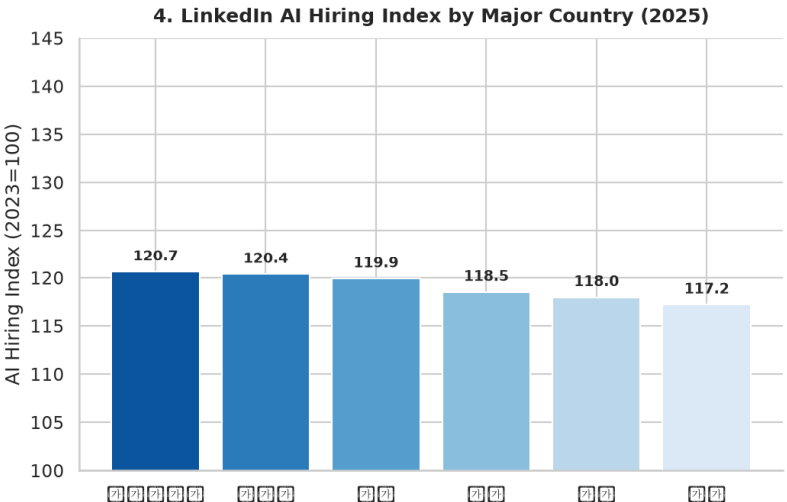
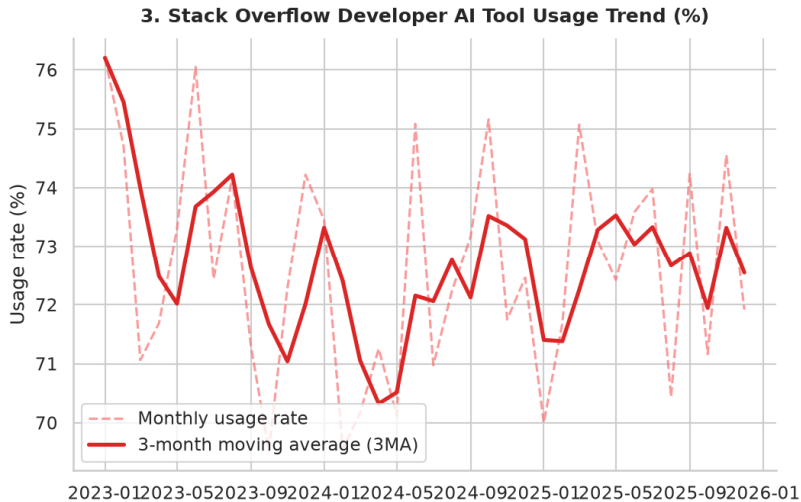
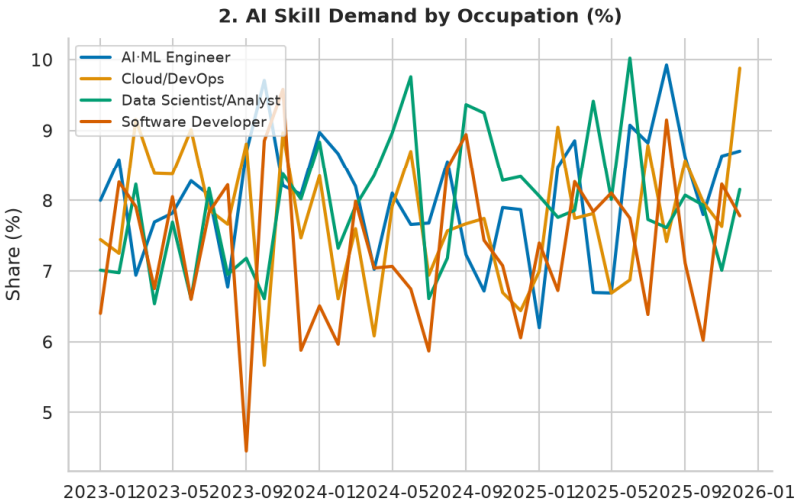
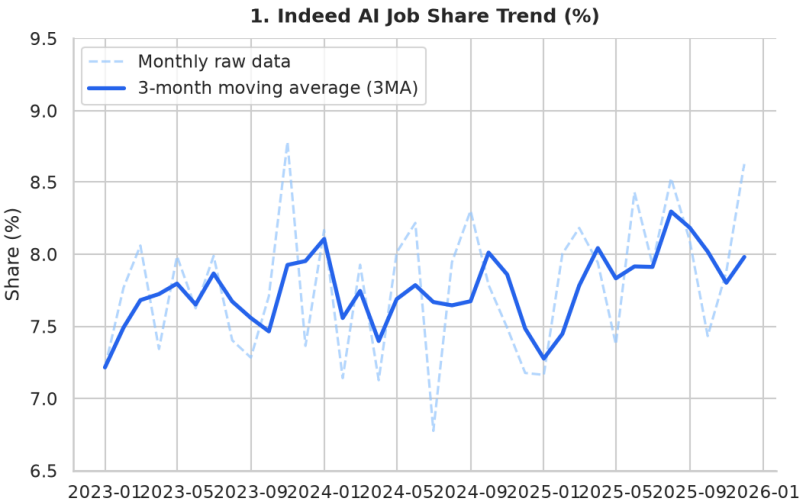
app.py

README.md

Outline

Timeline

Generative AI Adoption and Rising AI Skill Demand in Global IT Labor Markets (2023-2025)



Source: Indeed Hiring Lab, Stack Overflow Developer Survey, LinkedIn Economic Graph (2023-2025 combined dataset)

# AI Trend Analysis Dashboard

Interactive analysis of AI hiring demand, technology adoption, and labor market signals

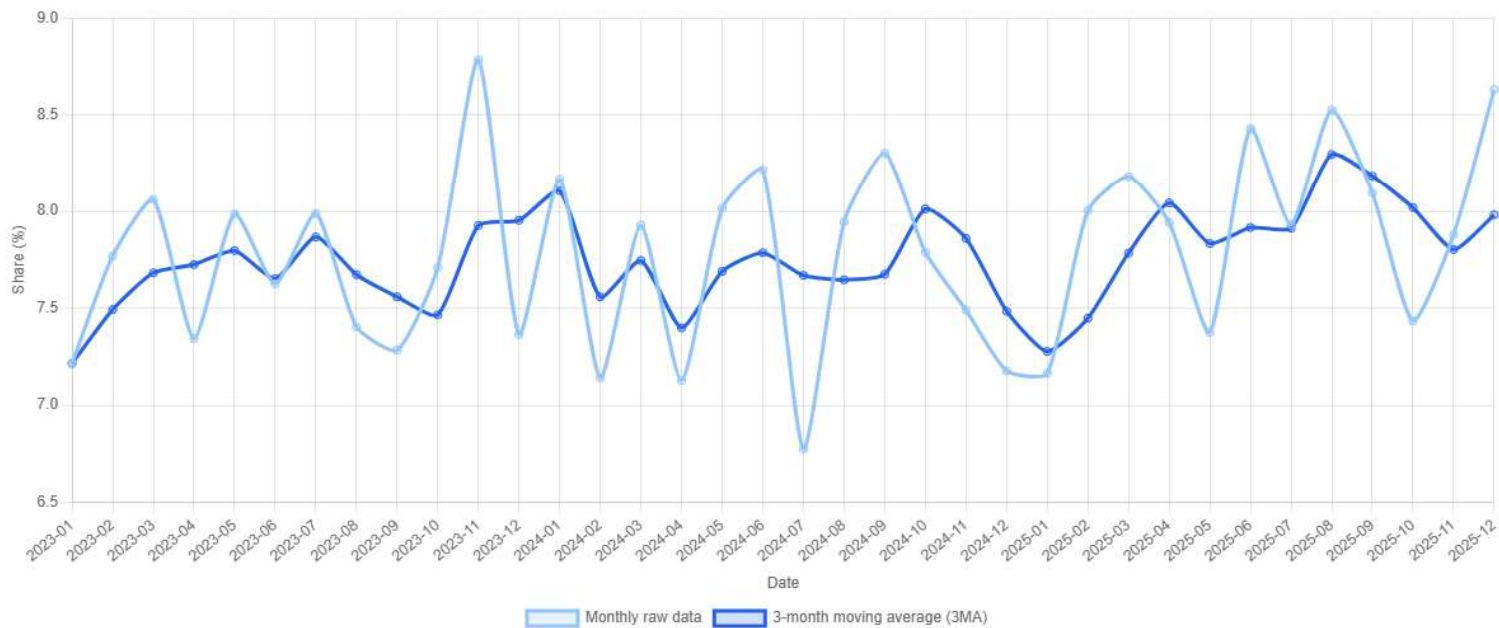
Indeed Trend

Occupation

Stack Overflow

Country

## Indeed AI Job Share Trend (%)



| Date    | Monthly raw data | 3-month moving average (3MA) |
|---------|------------------|------------------------------|
| 2023-01 | 7.22             | 7.22                         |
| 2023-02 | 7.77             | 7.49                         |

## AI Trend Analysis Dashboard

Interactive analysis of AI hiring demand, technology adoption, and labor market signals

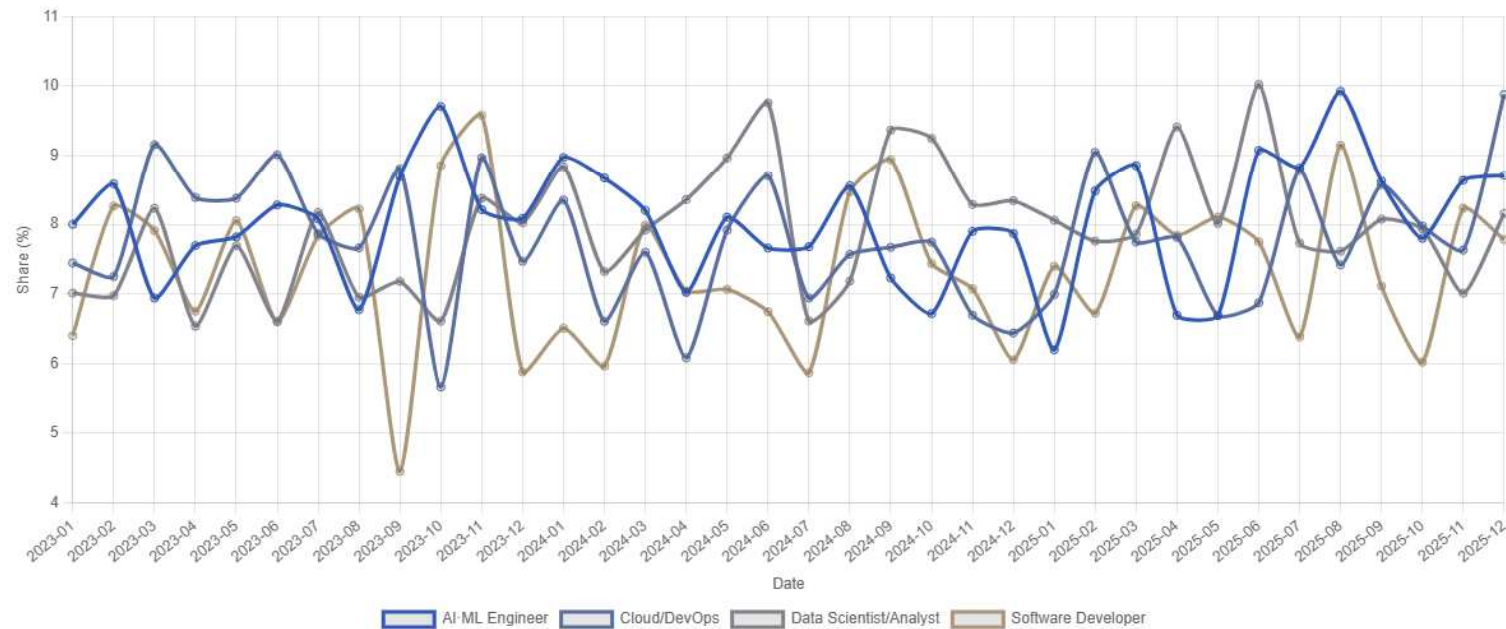
Indeed Trend

Occupation

Stack Overflow

Country

### AI Skill Demand by Occupation (%)



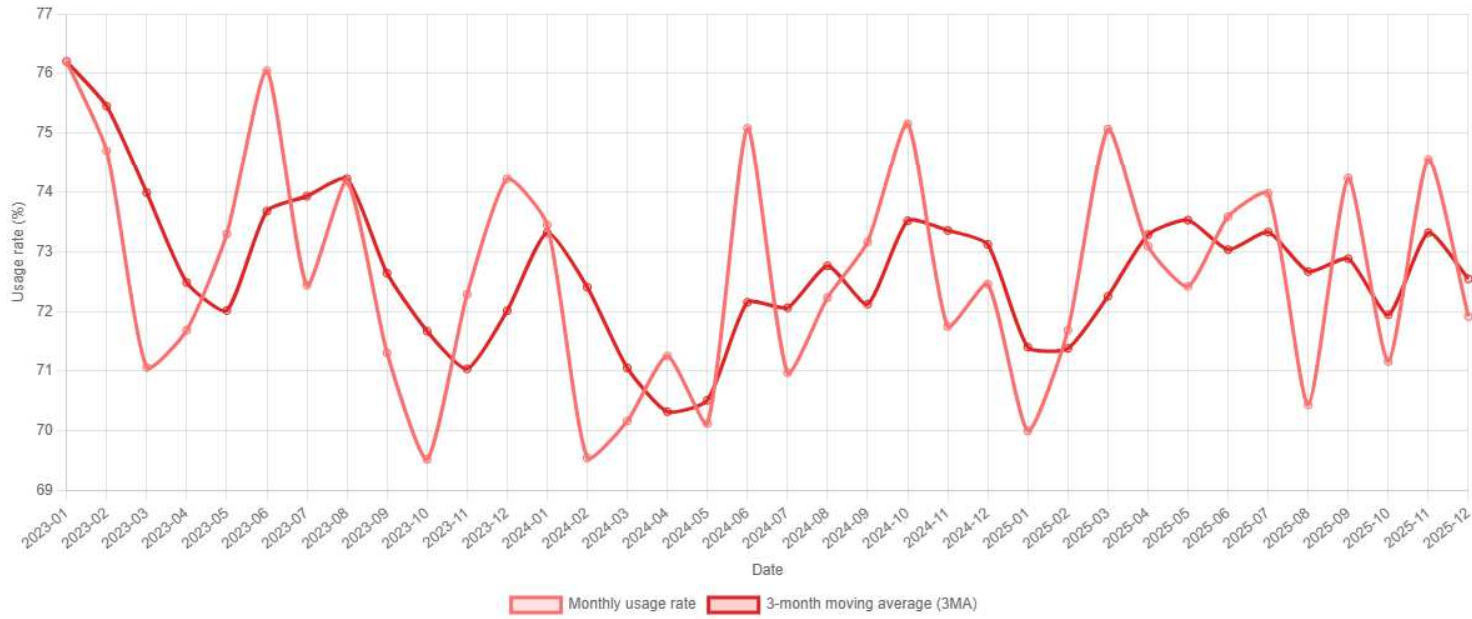
| Date    | AI-ML Engineer | Cloud/DevOps | Data Scientist/Analyst | Software Developer |
|---------|----------------|--------------|------------------------|--------------------|
| 2023-01 | 8              | 7.45         | 7.01                   | 6.4                |
| 2023-02 | 8.58           | 7.25         | 6.98                   | 8.26               |

# AI Trend Analysis Dashboard

Interactive analysis of AI hiring demand, technology adoption, and labor market signals

Indeed Trend | Occupation | **Stack Overflow** | Country

Stack Overflow Developer AI Tool Usage Trend (%)



| Date    | Monthly usage rate | 3-month moving average (3MA) |
|---------|--------------------|------------------------------|
| 2023-01 | 76.2               | 76.2                         |
| 2023-02 | 74.7               | 75.45                        |

